HW401 Instructor Transcript Guide

Slide-by-slide facilitation script with live internet demos

Built from Week 1 Day 1 through Week 3 Day 2 deck sequence
Includes the updated Week 3 Day 1 deck (ALFIE changed)

How to use this guide	For every slide, use the talk track as your base narrative, then open the suggested live demo or browse path, and finish with the delegate action prompt to keep the room active.

Facilitation note: Keep live browsing high level and analytical. Demonstrate doctrine, incident evidence, visualizations, and publicly available reference resources. Do not demonstrate offensive procedures, exploit execution, or operational tradecraft.

Live demo sources used throughout the guide

NATO cyber defence: https://www.nato.int/en/what-we-do/deterrence-and-defence/cyber-defence

Warsaw Summit Communique: https://www.nato.int/en/about-us/official-texts-and-resources/official-texts/2016/07/09/warsaw-summit-communique

NATO CCDCOE: https://ccdcoe.org/

MITRE ATT&CK Enterprise Matrix: https://attack.mitre.org/matrices/enterprise/

MITRE ATT&CK Navigator resources: https://attack.mitre.org/resources/attack-data-and-tools/

TeleGeography Submarine Cable Map: https://www.submarinecablemap.com/

Cloudflare Radar: https://radar.cloudflare.com/

Cloudflare Radar Outage Center: https://radar.cloudflare.com/outage-center

Internet Archive Wayback Machine: https://web.archive.org/

CISA Industrial Control Systems: https://www.cisa.gov/topics/industrial-control-systems

CISA ICS training catalogue: https://www.cisa.gov/resources-tools/programs/ics-training-available-through-cisa

EUvsDisinfo: https://euvsdisinfo.eu/

CISA deepfake threats advisory: https://www.cisa.gov/news-events/alerts/2023/09/12/nsa-fbi-and-cisa-release-cybersecurity-information-sheet-deepfake-threats

Week 1 Day 1

Deck file: HW401_Week1_Day1_Presentation.pptx | 20 slides

Use this section as your live instructor script. Each slide entry gives you a concise speaking flow, one internet-based illustration or demo path, and one delegate interaction or action to keep the room engaged.

#	Slide title	Primary facilitation angle
1	Introduction to Warfare Domains	Welcome the class, position the course, and frame cyberspace as the newest warfighting domain.
2	Lesson Outline	Walk the class through the lesson arc from start to finish so they understand the narrative structure.
3	Learning Objectives	Read the objectives as performance outcomes, not administrative text.
4	The Five Domains of Warfare	Introduce the domains as distinct operational spaces with different primary assets, but stress that modern conflict rarely stays inside one domain.
5	Land Domain	Use land as the baseline because it is the domain students intuitively understand.
6	Sea Domain	Frame sea power around movement, trade protection, and global reach.
7	Air Domain	Describe air power as speed, ISR, and strike reach.
8	Space Domain	Explain space as the enabling high ground that powers communications, navigation, timing, and surveillance.
9	Cyberspace Domain	Explain space as the enabling high ground that powers communications, navigation, timing, and surveillance.
10	Five Domains Comparison	Introduce the domains as distinct operational spaces with different primary assets, but stress that modern conflict rarely stays inside one domain.
11	Multi-Domain Operations	Explain that multi-domain operations are about synchronized effects, not separate teams working in parallel.
12	Cyberspace as an Operational Domain	Make the doctrinal argument clearly:

		cyberspace matters because military operations depend on it, adversaries can attack through it without physical proximity, and strategic effects can be achieved there.
13	Why Cyberspace is an Operational Domain	Make the doctrinal argument clearly: cyberspace matters because military operations depend on it, adversaries can attack through it without physical proximity, and strategic effects can be achieved there.
14	Key Characteristics of Cyber Warfare	Use these characteristics to show why cyber challenges traditional military assumptions.
15	Characteristics of Cyber Warfare	Use these characteristics to show why cyber challenges traditional military assumptions.
16	Borderless / Global Reach	Use these characteristics to show why cyber challenges traditional military assumptions.
17	Attribution Difficulty	Use these characteristics to show why cyber challenges traditional military assumptions.
18	Speed and Asymmetry	Use these characteristics to show why cyber challenges traditional military assumptions.
19	Key Takeaways	Use this slide to compress the lesson into five memory anchors.
20	Questions?	Use this slide as a deliberate debrief rather than a passive ending.

Slide 1. Introduction to Warfare Domains

On-slide cues: History and Trends of Warfare in the Fifth Domain

Instructor talk track	Welcome the class, position the course, and frame cyberspace as the newest warfighting domain. Tell delegates this module is not just about hacking; it is about how military, political, and information power now converge. Set the expectation that every later case study will connect back to these foundational concepts.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 2. Lesson Outline

On-slide cues: Course Overview and Semester Roadmap The Five Domains of Warfare Evolution of Warfare Domains Cyberspace as an Operational Domain Key Characteristics of Cyber Warfare

Instructor talk track	Walk the class through the lesson arc from start to finish so they understand the narrative structure. Emphasize that each section builds toward an analytical framework rather than a list of isolated facts. Use the outline to show where you will define concepts, examine cases, and then apply them.
Live internet demo / browse path	Browse https://www.cisa.gov/topics/industrial-control-systems and show the delegates how modern ICS and OT security is treated as national critical infrastructure security, not just traditional IT.
Delegate action / discussion move	Use the outline to set pace expectations and tell students where you want questions held for discussion versus asked immediately.

Slide 3. Learning Objectives

On-slide cues: Articulate the scope and expectations of HW401 Define the five domains of warfare Explain why cyberspace is an operational domain Identify key characteristics that distinguish cyber warfare

Instructor talk track	Read the objectives as performance outcomes, not administrative text. Tell delegates that by the end of the session they must be able to explain, compare, and apply these concepts under discussion pressure. This helps convert passive listening into active analysis from the start.
Live internet demo / browse path	Browse https://www.cisa.gov/topics/industrial-control-systems and show the delegates how modern ICS and OT security is treated as national critical infrastructure security, not just traditional IT.
Delegate action / discussion move	Tell delegates exactly how they will prove mastery by the end of the session: define, compare, analyse, and apply.

Slide 4. The Five Domains of Warfare

On-slide cues: Land, Sea, Air, Space, Cyberspace

Instructor talk	Introduce the domains as distinct operational spaces with different primary

track	assets, but stress that modern conflict rarely stays inside one domain. Explain that the slide gives students a framework for comparing how force, reach, control, and strategic advantage differ across land, sea, air, space, and cyber.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 5. Land Domain

On-slide cues: The oldest domain of warfare Control of territory, populations, resources Ground forces: infantry, armor, artillery Strategic value: Physical occupation

Instructor talk track	Use land as the baseline because it is the domain students intuitively understand. Explain occupation, terrain, and control of people and resources, then contrast that with cyber where presence is not required to generate strategic effect.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 6. Sea Domain

On-slide cues: Naval power and maritime control Protection of sea lanes and trade routes Power projection via carrier battle groups Strategic value: Global reach and logistics

Instructor talk track	Frame sea power around movement, trade protection, and global reach. Then make the modern connection: naval capability still depends on networks, satellite support, and undersea communications infrastructure, so maritime superiority now has an embedded cyber dependency.
Live internet demo / browse path	Browse https://www.cisa.gov/topics/industrial-control-systems and show the delegates how modern ICS and OT security is treated as national critical

	infrastructure security, not just traditional IT.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 7. Air Domain

On-slide cues: Aerial superiority and rapid deployment Intelligence, Surveillance, Reconnaissance Strategic bombing and close air support Strategic value: Speed and flexibility

Instructor talk track	Describe air power as speed, ISR, and strike reach. Then explain that modern air power is deeply digital, from mission planning to navigation, targeting, and aircraft systems, which means cyber disruption can shape air outcomes without direct dogfighting.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 8. Space Domain

On-slide cues: Satellites for communications, GPS, ISR Critical enabler for all other domains Anti-satellite weapons and space control Strategic value: Information superiority

Instructor talk track	Explain space as the enabling high ground that powers communications, navigation, timing, and surveillance. Stress that the strategic value of space is not only where the satellite is, but what all other domains lose when those services are denied or degraded.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 9. Cyberspace Domain

On-slide cues: Networks, data, and information systems Non-physical but operationally critical Formally recognized by NATO in 2016 Strategic value: Information advantage

Instructor talk track	Explain space as the enabling high ground that powers communications, navigation, timing, and surveillance. Stress that the strategic value of space is not only where the satellite is, but what all other domains lose when those services are denied or degraded.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 10. Five Domains Comparison

Instructor talk track	Introduce the domains as distinct operational spaces with different primary assets, but stress that modern conflict rarely stays inside one domain. Explain that the slide gives students a framework for comparing how force, reach, control, and strategic advantage differ across land, sea, air, space, and cyber.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Ask delegates to summarize the slide using the formula: concept, risk, and military implication.

Slide 11. Multi-Domain Operations

On-slide cues: Integration across all five domains Synchronized effects for advantage No domain operates in isolation Cyber enables and supports all domains

Instructor talk track	Explain that multi-domain operations are about synchronized effects, not separate teams working in parallel. The key teaching point is that cyber is both a warfighting domain and an enabler that connects command, intelligence, precision, and tempo across the other domains.

Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 12. Cyberspace as an Operational Domain

On-slide cues: Why the Fifth Domain Matters

Instructor talk track	Make the doctrinal argument clearly: cyberspace matters because military operations depend on it, adversaries can attack through it without physical proximity, and strategic effects can be achieved there. Tie the slide back to formal recognition by NATO and to the practical consequences for training, force design, and collective defence.
Live internet demo / browse path	Open https://www.nato.int/en/what-we-do/deterrence-and-defence/cyber-defence or the Warsaw communique at https://www.nato.int/en/about-us/official-texts-and-resources/official-texts/2016/07/09/warsaw-summit-communique. Read the exact wording that recognizes cyberspace as a domain of operations and ask delegates what that changes for force planning.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 13. Why Cyberspace is an Operational Domain

On-slide cues: Military operations depend on cyberspace Adversaries can attack without physical presence Critical infrastructure is digitally connected Information advantage determines outcomes Every soldier carries cyber-dependent equipment

Instructor talk track	Make the doctrinal argument clearly: cyberspace matters because military operations depend on it, adversaries can attack through it without physical proximity, and strategic effects can be achieved there. Tie the slide back to formal recognition by NATO and to the practical consequences for training, force design, and collective defence.
Live internet	Open https://www.nato.int/en/what-we-do/deterrence-and-defence/cyber-

demo / browse path	defence or the Warsaw communique at https://www.nato.int/en/about-us/official-texts-and-resources/official-texts/2016/07/09/warsaw-summit-communique. Read the exact wording that recognizes cyberspace as a domain of operations and ask delegates what that changes for force planning.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 14. Key Characteristics of Cyber Warfare

On-slide cues: What Makes Cyberspace Different

Instructor talk track	Use these characteristics to show why cyber challenges traditional military assumptions. Borderlessness affects defence, attribution affects deterrence, speed compresses decision time, and asymmetry lets weaker actors impose outsized costs. Keep translating each characteristic into a problem for commanders and policymakers.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 15. Characteristics of Cyber Warfare

On-slide cues: Borderless - Global reach Stealthy - Undetected operations Attribution Difficulty Speed - Near-instantaneous effects Asymmetric - Small vs large powers Less-Lethal - Often non-kinetic

Instructor talk track	Use these characteristics to show why cyber challenges traditional military assumptions. Borderlessness affects defence, attribution affects deterrence, speed compresses decision time, and asymmetry lets weaker actors impose outsized costs. Keep translating each characteristic into a problem for commanders and policymakers.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and

	cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 16. Borderless / Global Reach

On-slide cues: Cyber operations transcend physical boundaries Attack from anywhere to anywhere instantly No need for physical presence near target Traditional defenses cannot intercept Example: Adversary in one country targets systems globally

Instructor talk track	Use these characteristics to show why cyber challenges traditional military assumptions. Borderlessness affects defence, attribution affects deterrence, speed compresses decision time, and asymmetry lets weaker actors impose outsized costs. Keep translating each characteristic into a problem for commanders and policymakers.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 17. Attribution Difficulty

On-slide cues: Identifying the true source is challenging Attackers use proxy servers, VPNs, compromised systems False flags can mimic other actors Attribution possible but requires time and intelligence Complicates response and deterrence

Instructor talk track	Use these characteristics to show why cyber challenges traditional military assumptions. Borderlessness affects defence, attribution affects deterrence, speed compresses decision time, and asymmetry lets weaker actors impose outsized costs. Keep translating each characteristic into a problem for commanders and policymakers.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.

Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 18. Speed and Asymmetry

On-slide cues: Speed: Effects occur at near-light speed Malware can spread globally in hours Asymmetry: Small actors vs. large powers Criminal groups can threaten corporations Small nations can challenge major powers Low cost of entry compared to traditional warfare

Instructor talk track	Use these characteristics to show why cyber challenges traditional military assumptions. Borderlessness affects defence, attribution affects deterrence, speed compresses decision time, and asymmetry lets weaker actors impose outsized costs. Keep translating each characteristic into a problem for commanders and policymakers.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 19. Key Takeaways

On-slide cues: Five domains: Land, Sea, Air, Space, Cyberspace Cyberspace recognized as operational domain (NATO 2016) Multi-domain operations integrate all five domains Cyber warfare characteristics: borderless, stealthy, attribution difficulty, speed, asymmetry, less-lethal

Instructor talk track	Use this slide to compress the lesson into five memory anchors. Re-state each point in plain language, then translate it into why it matters for commanders, analysts, or defenders. This is the moment to check whether students can move from vocabulary to judgement.
Live internet demo / browse path	Browse https://www.cisa.gov/topics/industrial-control-systems and show the delegates how modern ICS and OT security is treated as national critical infrastructure security, not just traditional IT.
Delegate action / discussion move	Ask delegates to summarize the slide using the formula: concept, risk, and military implication.

Slide 20. Questions?

Instructor talk track	Use this slide as a deliberate debrief rather than a passive ending. Summarize the one idea students must carry forward from the session, restate the operational significance, and then invite analytical questions, not just clarification questions. Close by linking the current session to the next day so the storyline feels continuous.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Invite one operational question, one ethical question, and one technical question before closing.

Week 1 Day 2

Deck file: HW401_Week1_Day2_Presentation.pptx | 21 slides

Use this section as your live instructor script. Each slide entry gives you a concise speaking flow, one internet-based illustration or demo path, and one delegate interaction or action to keep the room engaged.

#	Slide title	Primary facilitation angle
1	Domains Deep-Dive	Use this slide to explain the operational meaning of the listed points: History and Trends of Warfare in the Fifth Domain.
2	Lesson Outline	Walk the class through the lesson arc from start to finish so they understand the narrative structure.
3	Learning Objectives	Read the objectives as performance outcomes, not administrative text.
4	Evolution of Warfare Domains	Teach this slide as a timeline of adaptation.
5	Historical Evolution	Teach this slide as a timeline of adaptation.
6	Key Milestones	Teach this slide as a timeline of adaptation.
7	The Three Layers of Cyberspace	Explain the layers as a disciplined way to analyse where an operation acts and where it creates consequences.
8	Physical Layer	Explain the layers as a disciplined way to analyse where an operation acts and where it creates consequences.
9	Logical Layer	Explain the layers as a disciplined way to analyse where an operation acts and where it creates consequences.
10	Cognitive Layer	Explain the layers as a disciplined way to analyse where an operation acts and where it creates consequences.
11	Three Layers Summary	Explain the layers as a disciplined way to analyse where an operation acts and where it creates consequences.
12	Multi-Domain Operations	Explain that multi-domain operations are about synchronized effects, not separate teams working in parallel.
13	Multi-Domain Operations (MDO)	Explain that multi-domain operations are about synchronized effects, not separate

		teams working in parallel.
14	Cyber's Role in MDO	Use this slide to explain the operational meaning of the listed points: Intelligence gathering across all domains Command and control communications GPS and navigation (Space-Cyber integration) Weapons systems increasingly networked Information operations support Electronic warfare coordination.
15	Critical Infrastructure	Frame critical infrastructure as the set of services whose disruption creates cascading national consequences.
16	Critical Infrastructure Sectors	Frame critical infrastructure as the set of services whose disruption creates cascading national consequences.
17	The Dual-Use Challenge	Frame critical infrastructure as the set of services whose disruption creates cascading national consequences.
18	NATO and Cyberspace	Use this slide to explain the operational meaning of the listed points: 2016: Cyberspace declared operational domain Article 5 can apply to cyber attacks Collective defence in cyberspace NATO CCDCOE in Tallinn, Estonia Tallinn Manual on cyber international law.
19	Key Takeaways	Use this slide to compress the lesson into five memory anchors.
20	Preview: Day 3 Lab	Position the next class as a continuation of today’s logic.
21	Questions?	Use this slide as a deliberate debrief rather than a passive ending.

Slide 1. Domains Deep-Dive

On-slide cues: History and Trends of Warfare in the Fifth Domain

Instructor talk track	Use this slide to explain the operational meaning of the listed points: History and Trends of Warfare in the Fifth Domain. Do not just read the bullets. Translate them into what a commander, analyst, policymaker, or defender should notice, care about, and do differently.
Live internet	Use a quick live browse to anchor one current, real-world example connected

demo / browse path	to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 2. Lesson Outline

On-slide cues: Evolution of Warfare Domains The Three Layers of Cyberspace Multi-Domain Operations (MDO) The Information Environment Critical Infrastructure and Cyberspace

Instructor talk track	Walk the class through the lesson arc from start to finish so they understand the narrative structure. Emphasize that each section builds toward an analytical framework rather than a list of isolated facts. Use the outline to show where you will define concepts, examine cases, and then apply them.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Use the outline to set pace expectations and tell students where you want questions held for discussion versus asked immediately.

Slide 3. Learning Objectives

On-slide cues: Trace the historical evolution of warfare domains Describe the three layers of cyberspace Explain multi-domain operations concept Identify critical infrastructure dependencies Connect cyberspace to the information environment

Instructor talk track	Read the objectives as performance outcomes, not administrative text. Tell delegates that by the end of the session they must be able to explain, compare, and apply these concepts under discussion pressure. This helps convert passive listening into active analysis from the start.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.

Delegate action / discussion move	Tell delegates exactly how they will prove mastery by the end of the session: define, compare, analyse, and apply.

Slide 4. Evolution of Warfare Domains

On-slide cues: From Ancient Battlefields to Cyberspace

Instructor talk track	Teach this slide as a timeline of adaptation. Each new domain became militarily meaningful when technology made sustained operations possible there. The purpose of the slide is to show students that cyber's rise is part of a long pattern of military change, not a complete historical break.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 5. Historical Evolution

On-slide cues: Ancient: Land warfare dominates 1800s: Naval power becomes strategic 1900s: Air power emerges (WWI/WWII) 1960s: Space operations begin 2010s: Cyberspace formally recognized

Instructor talk track	Teach this slide as a timeline of adaptation. Each new domain became militarily meaningful when technology made sustained operations possible there. The purpose of the slide is to show students that cyber's rise is part of a long pattern of military change, not a complete historical break.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 6. Key Milestones

On-slide cues: 1903: Wright Brothers - Aviation begins 1957: Sputnik - Space age begins 1969: ARPANET - Internet precursor 1988: Morris Worm - First major cyber incident 2009: US Cyber Command established 2016: NATO recognizes cyberspace as operational domain

Instructor talk track	Teach this slide as a timeline of adaptation. Each new domain became militarily meaningful when technology made sustained operations possible there. The purpose of the slide is to show students that cyber's rise is part of a long pattern of military change, not a complete historical break.
Live internet demo / browse path	Use https://web.archive.org/ to show delegates how older web content can still be surfaced for historical analysis. This helps students see that cyber history is recoverable evidence, not just storytelling.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 7. The Three Layers of Cyberspace

On-slide cues: Physical, Logical, Cognitive

Instructor talk track	Explain the layers as a disciplined way to analyse where an operation acts and where it creates consequences. The physical layer is infrastructure, the logical layer is software and protocols, and the cognitive layer is human interpretation and choice. Repeatedly stress that serious operations often chain effects across all three.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 8. Physical Layer

On-slide cues: Hardware and infrastructure Servers, routers, cables, data centers Fiber optic networks, satellites The tangible foundation of cyberspace Vulnerable to physical attack

Instructor talk	Explain the layers as a disciplined way to analyse where an operation acts

track	and where it creates consequences. The physical layer is infrastructure, the logical layer is software and protocols, and the cognitive layer is human interpretation and choice. Repeatedly stress that serious operations often chain effects across all three.
Live internet demo / browse path	Browse https://www.submarinecablemap.com/ and zoom into Africa or Europe. Show that cyberspace still depends on physical chokepoints, landing stations, and owned infrastructure.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 9. Logical Layer

On-slide cues: Software, protocols, and data Operating systems, applications Network protocols (TCP/IP, HTTP) Code that makes systems function Target of malware and exploitation

Instructor talk track	Explain the layers as a disciplined way to analyse where an operation acts and where it creates consequences. The physical layer is infrastructure, the logical layer is software and protocols, and the cognitive layer is human interpretation and choice. Repeatedly stress that serious operations often chain effects across all three.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 10. Cognitive Layer

On-slide cues: Human perception and decision-making How people interpret information Trust, belief, and understanding Target of influence operations Most complex layer to defend

Instructor talk track	Explain the layers as a disciplined way to analyse where an operation acts and where it creates consequences. The physical layer is infrastructure, the logical layer is software and protocols, and the cognitive layer is human interpretation and choice. Repeatedly stress that serious operations often

	chain effects across all three.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 11. Three Layers Summary

On-slide cues: Physical: Hardware, infrastructure (WHAT) Logical: Software, protocols, data (HOW) Cognitive: Human perception, decisions (WHO) Operations can target any or all layers Comprehensive defence must address all three

Instructor talk track	Explain the layers as a disciplined way to analyse where an operation acts and where it creates consequences. The physical layer is infrastructure, the logical layer is software and protocols, and the cognitive layer is human interpretation and choice. Repeatedly stress that serious operations often chain effects across all three.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Ask delegates to summarize the slide using the formula: concept, risk, and military implication.

Slide 12. Multi-Domain Operations

On-slide cues: Integrating the Five Domains

Instructor talk track	Explain that multi-domain operations are about synchronized effects, not separate teams working in parallel. The key teaching point is that cyber is both a warfighting domain and an enabler that connects command, intelligence, precision, and tempo across the other domains.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and

	cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 13. Multi-Domain Operations (MDO)

On-slide cues: Integration across all five domains Rapid combination of capabilities Creates dilemmas for adversaries No domain operates in isolation Cyber enables all other domains

Instructor talk track	Explain that multi-domain operations are about synchronized effects, not separate teams working in parallel. The key teaching point is that cyber is both a warfighting domain and an enabler that connects command, intelligence, precision, and tempo across the other domains.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 14. Cyber's Role in MDO

On-slide cues: Intelligence gathering across all domains Command and control communications GPS and navigation (Space-Cyber integration) Weapons systems increasingly networked Information operations support Electronic warfare coordination

Instructor talk track	Use this slide to explain the operational meaning of the listed points: Intelligence gathering across all domains Command and control communications GPS and navigation (Space-Cyber integration) Weapons systems increasingly networked Information operations support Electronic warfare coordination. Do not just read the bullets. Translate them into what a commander, analyst, policymaker, or defender should notice, care about, and do differently.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.

Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 15. Critical Infrastructure

On-slide cues: Cyberspace Dependencies

Instructor talk track	Frame critical infrastructure as the set of services whose disruption creates cascading national consequences. Then push the class to see the dual-use problem: civilian systems often carry military relevance, which complicates defence, deterrence, escalation, and legal targeting decisions.
Live internet demo / browse path	Browse https://www.cisa.gov/topics/industrial-control-systems and show the delegates how modern ICS and OT security is treated as national critical infrastructure security, not just traditional IT.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 16. Critical Infrastructure Sectors

On-slide cues: Energy (power plants, grid) Water and wastewater systems Transportation (air, rail, maritime) Communications (telecom, internet) Financial services Healthcare and emergency services

Instructor talk track	Frame critical infrastructure as the set of services whose disruption creates cascading national consequences. Then push the class to see the dual-use problem: civilian systems often carry military relevance, which complicates defence, deterrence, escalation, and legal targeting decisions.
Live internet demo / browse path	Browse https://www.cisa.gov/topics/industrial-control-systems and show the delegates how modern ICS and OT security is treated as national critical infrastructure security, not just traditional IT.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 17. The Dual-Use Challenge

On-slide cues: Military and civilian share same infrastructure Internet backbone serves both sectors Power grid powers military bases Commercial satellites support military comms Attack on civilian can affect military (and vice versa)

Instructor talk track	Frame critical infrastructure as the set of services whose disruption creates cascading national consequences. Then push the class to see the dual-use problem: civilian systems often carry military relevance, which complicates defence, deterrence, escalation, and legal targeting decisions.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 18. NATO and Cyberspace

On-slide cues: 2016: Cyberspace declared operational domain Article 5 can apply to cyber attacks Collective defence in cyberspace NATO CCDCOE in Tallinn, Estonia Tallinn Manual on cyber international law

Instructor talk track	Use this slide to explain the operational meaning of the listed points: 2016: Cyberspace declared operational domain Article 5 can apply to cyber attacks Collective defence in cyberspace NATO CCDCOE in Tallinn, Estonia Tallinn Manual on cyber international law. Do not just read the bullets. Translate them into what a commander, analyst, policymaker, or defender should notice, care about, and do differently.
Live internet demo / browse path	Open https://www.nato.int/en/what-we-do/deterrence-and-defence/cyber-defence or the Warsaw communique at https://www.nato.int/en/about-us/official-texts-and-resources/official-texts/2016/07/09/warsaw-summit-communique. Read the exact wording that recognizes cyberspace as a domain of operations and ask delegates what that changes for force planning.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 19. Key Takeaways

On-slide cues: Domains evolved: Land -> Sea -> Air -> Space -> Cyber Cyberspace has three layers: Physical, Logical, Cognitive Multi-Domain Operations integrate all domains Critical infrastructure is cyber-dependent NATO recognizes cyber as operational domain

Instructor talk track	Use this slide to compress the lesson into five memory anchors. Re-state each point in plain language, then translate it into why it matters for commanders, analysts, or defenders. This is the moment to check whether students can move from vocabulary to judgement.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Ask delegates to summarize the slide using the formula: concept, risk, and military implication.

Slide 20. Preview: Day 3 Lab

On-slide cues: Introduction to MITRE ATT&CK Framework Hands-on navigation of ATT&CK Matrix Threat group research assignment ATT&CK Navigator tool Mapping adversary techniques

Instructor talk track	Position the next class as a continuation of today's logic. Show delegates exactly how the next lesson will deepen or operationalize what they just learned so the course feels cumulative and coherent.
Live internet demo / browse path	Browse https://attack.mitre.org/matrices/enterprise/ and then launch the Navigator from https://attack.mitre.org/resources/attack-data-and-tools/. Show students how tactics, techniques, and heat maps will support the lab.
Delegate action / discussion move	Show them exactly what good output looks like so the next task feels executable, not vague.

Slide 21. Questions?

Instructor talk track	Use this slide as a deliberate debrief rather than a passive ending. Summarize the one idea students must carry forward from the session, restate the operational significance, and then invite analytical questions, not just clarification questions. Close by linking the current session to the next day so

	the storyline feels continuous.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Invite one operational question, one ethical question, and one technical question before closing.

Week 2 Day 1

Deck file: HW401_Week2_Day1_Presentation.pptx | 20 slides

Use this section as your live instructor script. Each slide entry gives you a concise speaking flow, one internet-based illustration or demo path, and one delegate interaction or action to keep the room engaged.

#	Slide title	Primary facilitation angle
1	Early Cyber Incidents	Use this slide to explain the operational meaning of the listed points: History and Trends of Warfare in the Fifth Domain.
2	Lesson Outline	Walk the class through the lesson arc from start to finish so they understand the narrative structure.
3	Learning Objectives	Read the objectives as performance outcomes, not administrative text.
4	The Morris Worm	Tell this story as the internet’s wake-up call.
5	The Morris Worm - Overview	Tell this story as the internet’s wake-up call.
6	How the Worm Worked	Use this slide to explain the operational meaning of the listed points: Exploited sendmail DEBUG command Used fingerd buffer overflow Guessed passwords via rsh/rexec Self-replicated across systems Bug: Re-infected already-infected systems.
7	Consequences and Response	Use this slide to explain the operational meaning of the listed points: CERT/CC Created at Carnegie Mellon First conviction under CFAA (1986) Demonstrated network interconnection risks Began coordinated vulnerability disclosure Morris: 3 years probation, now MIT professor.
8	The Cuckoo's Egg	Teach this as early proof that cyber espionage was already geopolitical before most institutions were ready to recognize it.
9	The Cliff Stoll Investigation	Teach this as early proof that cyber espionage was already geopolitical before most institutions were ready to recognize it.
10	Techniques Used (Pre-ATT&CK)	Use this slide to explain the operational meaning of the listed points: Password guessing (weak credentials) Exploiting trust between systems Privilege escalation Lateral

		movement across networks Data exfiltration via dial-up.
11	Moonlight Maze	Use this slide to show the maturation from isolated compromise to sustained state-backed espionage.
12	Moonlight Maze - Overview	Use this slide to show the maturation from isolated compromise to sustained state-backed espionage.
13	Strategic Implications	Use this slide to show the maturation from isolated compromise to sustained state-backed espionage.
14	Early Cyber Timeline	Use the timeline to show progression rather than isolated events.
15	Patterns and Lessons	Use this slide to show the maturation from isolated compromise to sustained state-backed espionage.
16	Common Themes Across Incidents	Use this slide to show the maturation from isolated compromise to sustained state-backed espionage.
17	Key Takeaways	Use this slide to compress the lesson into five memory anchors.
18	Preview: Day 2	Position the next class as a continuation of today's logic.
19	Homework Assignment	Explain the homework as an opportunity to practice analytical writing, not just incident description.
20	Questions?	Use this slide as a deliberate debrief rather than a passive ending.

Slide 1. Early Cyber Incidents

On-slide cues: History and Trends of Warfare in the Fifth Domain

Instructor talk track	Use this slide to explain the operational meaning of the listed points: History and Trends of Warfare in the Fifth Domain. Do not just read the bullets. Translate them into what a commander, analyst, policymaker, or defender should notice, care about, and do differently.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and

	cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 2. Lesson Outline

On-slide cues: The Morris Worm (1988) - The Wake-Up Call The Cuckoo's Egg - Early Espionage Moonlight Maze - Nation-State Emergence Patterns and Lessons Learned

Instructor talk track	Walk the class through the lesson arc from start to finish so they understand the narrative structure. Emphasize that each section builds toward an analytical framework rather than a list of isolated facts. Use the outline to show where you will define concepts, examine cases, and then apply them.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Use the outline to set pace expectations and tell students where you want questions held for discussion versus asked immediately.

Slide 3. Learning Objectives

On-slide cues: Describe the Morris Worm's significance Explain early espionage cases (Cuckoo's Egg) Analyze Moonlight Maze as state-sponsored espionage Identify patterns connecting early to modern incidents Explain policy/organizational responses to incidents

Instructor talk track	Read the objectives as performance outcomes, not administrative text. Tell delegates that by the end of the session they must be able to explain, compare, and apply these concepts under discussion pressure. This helps convert passive listening into active analysis from the start.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Tell delegates exactly how they will prove mastery by the end of the session: define, compare, analyse, and apply.

Slide 4. The Morris Worm

On-slide cues: November 2, 1988 - The Wake-Up Call

Instructor talk track	Tell this story as the internet's wake-up call. Emphasize that the lesson is not just the worm's code, but the systemic insight it revealed: interconnected systems can fail at scale from a single flawed operation, even when intent is not overtly destructive.
Live internet demo / browse path	Use https://web.archive.org/ to show delegates how older web content can still be surfaced for historical analysis. This helps students see that cyber history is recoverable evidence, not just storytelling.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 5. The Morris Worm - Overview

On-slide cues: Date: November 2, 1988 Creator: Robert Tappan Morris (Cornell) Target: Unix systems on ARPANET Impact: ~6,000 systems (10% of internet) Intent: Experimental, not malicious

Instructor talk track	Tell this story as the internet's wake-up call. Emphasize that the lesson is not just the worm's code, but the systemic insight it revealed: interconnected systems can fail at scale from a single flawed operation, even when intent is not overtly destructive.
Live internet demo / browse path	Use https://web.archive.org/ to show delegates how older web content can still be surfaced for historical analysis. This helps students see that cyber history is recoverable evidence, not just storytelling.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 6. How the Worm Worked

On-slide cues: Exploited sendmail DEBUG command Used fingerd buffer overflow Guessed passwords via rsh/rexec Self-replicated across systems Bug: Re-infected already-infected systems

Instructor talk track	Use this slide to explain the operational meaning of the listed points: Exploited sendmail DEBUG command Used fingerd buffer overflow Guessed

	passwords via rsh/rexec Self-replicated across systems Bug: Re-infected already-infected systems. Do not just read the bullets. Translate them into what a commander, analyst, policymaker, or defender should notice, care about, and do differently.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 7. Consequences and Response

On-slide cues: CERT/CC Created at Carnegie Mellon First conviction under CFAA (1986) Demonstrated network interconnection risks Began coordinated vulnerability disclosure Morris: 3 years probation, now MIT professor

Instructor talk track	Use this slide to explain the operational meaning of the listed points: CERT/CC Created at Carnegie Mellon First conviction under CFAA (1986) Demonstrated network interconnection risks Began coordinated vulnerability disclosure Morris: 3 years probation, now MIT professor. Do not just read the bullets. Translate them into what a commander, analyst, policymaker, or defender should notice, care about, and do differently.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 8. The Cuckoo's Egg

On-slide cues: 1986-1989: Early Cyber Espionage

Instructor talk track	Teach this as early proof that cyber espionage was already geopolitical before most institutions were ready to recognize it. Highlight the patience of the investigation, the value of anomaly detection, and the enduring relevance of

	credential theft, lateral movement, and weak trust boundaries.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 9. The Cliff Stoll Investigation

On-slide cues: 1986: Stoll notices 75-cent billing error Traces unauthorized access to Germany Hacker Markus Hess stealing military data Connection discovered: Hess selling to KGB Investigation lasted nearly one year

Instructor talk track	Teach this as early proof that cyber espionage was already geopolitical before most institutions were ready to recognize it. Highlight the patience of the investigation, the value of anomaly detection, and the enduring relevance of credential theft, lateral movement, and weak trust boundaries.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 10. Techniques Used (Pre-ATT&CK)

On-slide cues: Password guessing (weak credentials) Exploiting trust between systems Privilege escalation Lateral movement across networks Data exfiltration via dial-up

Instructor talk track	Use this slide to explain the operational meaning of the listed points: Password guessing (weak credentials) Exploiting trust between systems Privilege escalation Lateral movement across networks Data exfiltration via dial-up. Do not just read the bullets. Translate them into what a commander, analyst, policymaker, or defender should notice, care about, and do differently.
Live internet	Browse https://attack.mitre.org/matrices/enterprise/ and then launch the

demo / browse path	Navigator from https://attack.mitre.org/resources/attack-data-and-tools/. Show students how tactics, techniques, and heat maps will support the lab.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 11. Moonlight Maze

On-slide cues: 1998: Nation-State Espionage Emerges

Instructor talk track	Use this slide to show the maturation from isolated compromise to sustained state-backed espionage. The key teaching point is continuity: many modern APT patterns were visible early, and policy responses repeatedly lagged behind operational innovation.
Live internet demo / browse path	Use https://web.archive.org/ to show delegates how older web content can still be surfaced for historical analysis. This helps students see that cyber history is recoverable evidence, not just storytelling.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 12. Moonlight Maze - Overview

On-slide cues: Discovered: 1998 (began ~1996) Attribution: Russia (never officially confirmed) Targets: Pentagon, NASA, DoE, universities Duration: At least 2 years sustained access Scale: Data 'taller than Washington Monument'

Instructor talk track	Use this slide to show the maturation from isolated compromise to sustained state-backed espionage. The key teaching point is continuity: many modern APT patterns were visible early, and policy responses repeatedly lagged behind operational innovation.
Live internet demo / browse path	Use https://web.archive.org/ to show delegates how older web content can still be surfaced for historical analysis. This helps students see that cyber history is recoverable evidence, not just storytelling.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 13. Strategic Implications

On-slide cues: Proof of concept for nation-state cyber ops Asymmetric advantage for economically weaker nations Prompted PDD-63 (critical infrastructure policy) Precursor to modern APT groups Possible link to modern Turla APT

Instructor talk track	Use this slide to show the maturation from isolated compromise to sustained state-backed espionage. The key teaching point is continuity: many modern APT patterns were visible early, and policy responses repeatedly lagged behind operational innovation.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 14. Early Cyber Timeline

On-slide cues: 1986: Cuckoo's Egg begins (KGB espionage) 1988: Morris Worm -> CERT created 1996: Moonlight Maze begins 1998: Moonlight Maze discovered 1998: PDD-63 (Policy response)

Instructor talk track	Use the timeline to show progression rather than isolated events. The slide works best when you narrate the shift from experimentation to espionage, then disruption, hybrid integration, and finally physical destruction.
Live internet demo / browse path	Use https://web.archive.org/ to show delegates how older web content can still be surfaced for historical analysis. This helps students see that cyber history is recoverable evidence, not just storytelling.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 15. Patterns and Lessons

On-slide cues: Connecting the Dots

Instructor talk track	Use this slide to show the maturation from isolated compromise to sustained state-backed espionage. The key teaching point is continuity: many modern

	APT patterns were visible early, and policy responses repeatedly lagged behind operational innovation.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 16. Common Themes Across Incidents

On-slide cues: Exploited trust (credentials, connections) Cascading effects across systems Response gaps (no jurisdiction, no playbook) Attribution challenges (months to years) Policy always follows, never leads

Instructor talk track	Use this slide to show the maturation from isolated compromise to sustained state-backed espionage. The key teaching point is continuity: many modern APT patterns were visible early, and policy responses repeatedly lagged behind operational innovation.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 17. Key Takeaways

On-slide cues: Morris Worm: Interconnected vulnerability -> CERT Cuckoo's Egg: Attribution possible but difficult Moonlight Maze: State-sponsored espionage at scale Pattern: Incident -> Reaction -> Policy (behind)

Instructor talk track	Use this slide to compress the lesson into five memory anchors. Re-state each point in plain language, then translate it into why it matters for commanders, analysts, or defenders. This is the moment to check whether students can move from vocabulary to judgement.
Live internet	Use a quick live browse to anchor one current, real-world example connected

demo / browse path	to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Ask delegates to summarize the slide using the formula: concept, risk, and military implication.

Slide 18. Preview: Day 2

On-slide cues: Titan Rain (2003-2007): Sustained Chinese espionage Estonia (2007): First cyber attack on nation-state Georgia (2008): Cyber integrated with kinetic ops Stuxnet (2010): The first true cyber weapon Escalation: Espionage -> Disruption -> Destruction

Instructor talk track	Position the next class as a continuation of today's logic. Show delegates exactly how the next lesson will deepen or operationalize what they just learned so the course feels cumulative and coherent.
Live internet demo / browse path	Live browse a reputable search result for a public explainer on GPS jamming or radar spoofing, then contrast electromagnetic attack with pure network intrusion. Keep the demo conceptual and avoid any step-by-step tradecraft.
Delegate action / discussion move	Show them exactly what good output looks like so the next task feels executable, not vague.

Slide 19. Homework Assignment

On-slide cues: Research ONE 1990s cyber incident (not covered today) Write 200-word summary covering: - What happened - Why it was significant - What response/policy resulted Due: Day 4

Instructor talk track	Explain the homework as an opportunity to practice analytical writing, not just incident description. Tell delegates you want significance, policy consequence, and why the case still matters today.
Live internet demo / browse path	Live browse a reputable search result for a public explainer on GPS jamming or radar spoofing, then contrast electromagnetic attack with pure network intrusion. Keep the demo conceptual and avoid any step-by-step tradecraft.
Delegate action / discussion move	Show them exactly what good output looks like so the next task feels executable, not vague.

Slide 20. Questions?

Instructor talk track	Use this slide as a deliberate debrief rather than a passive ending. Summarize the one idea students must carry forward from the session, restate the operational significance, and then invite analytical questions, not just clarification questions. Close by linking the current session to the next day so the storyline feels continuous.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Invite one operational question, one ethical question, and one technical question before closing.

Week 2 Day 2

Deck file: HW401_Week2_Day2_Presentation.pptx | 24 slides

Use this section as your live instructor script. Each slide entry gives you a concise speaking flow, one internet-based illustration or demo path, and one delegate interaction or action to keep the room engaged.

#	Slide title	Primary facilitation angle
1	Major Cyber Operations 2000-2010	Use this slide to explain the operational meaning of the listed points: History and Trends of Warfare in the Fifth Domain.
2	Lesson Outline	Walk the class through the lesson arc from start to finish so they understand the narrative structure.
3	Learning Objectives	Read the objectives as performance outcomes, not administrative text.
4	Cyber History Timeline	Use the timeline to show progression rather than isolated events.
5	Titan Rain	Frame Titan Rain as strategic espionage at scale.
6	Titan Rain - Overview	Frame Titan Rain as strategic espionage at scale.
7	Strategic Implications	Use this slide to show the maturation from isolated compromise to sustained state-backed espionage.
8	Estonia 2007	Teach Estonia as a national-level disruption case where the technical attacks were less important than the political and cognitive effects.
9	The Bronze Soldier Crisis	Teach Estonia as a national-level disruption case where the technical attacks were less important than the political and cognitive effects.
10	The Cyber Campaign	Teach Estonia as a national-level disruption case where the technical attacks were less important than the political and cognitive effects.
11	Why Estonia Mattered	Teach Estonia as a national-level disruption case where the technical attacks were less important than the political and cognitive

		effects.
12	Georgia 2008	Use Georgia to illustrate cyber-kinetic integration.
13	The Five-Day War	Use Georgia to illustrate cyber-kinetic integration.
14	What Made Georgia Different	Use Georgia to illustrate cyber-kinetic integration.
15	Lessons for Military Doctrine	Use Georgia to illustrate cyber-kinetic integration.
16	Stuxnet	Teach Stuxnet as the moment cyber crossed decisively into controlled physical sabotage.
17	Stuxnet - Discovery	Teach Stuxnet as the moment cyber crossed decisively into controlled physical sabotage.
18	How Stuxnet Worked	Teach Stuxnet as the moment cyber crossed decisively into controlled physical sabotage.
19	Physical Destruction via Cyber	Teach Stuxnet as the moment cyber crossed decisively into controlled physical sabotage.
20	Implications and Legacy	Teach Stuxnet as the moment cyber crossed decisively into controlled physical sabotage.
21	The Escalation Pattern	Use the timeline to show progression rather than isolated events.
22	Key Takeaways	Use this slide to compress the lesson into five memory anchors.
23	Preview: Day 3 Lab	Position the next class as a continuation of today's logic.
24	Questions?	Use this slide as a deliberate debrief rather than a passive ending.

Slide 1. Major Cyber Operations 2000-2010

On-slide cues: History and Trends of Warfare in the Fifth Domain

Instructor talk track	Use this slide to explain the operational meaning of the listed points: History and Trends of Warfare in the Fifth Domain. Do not just read the bullets. Translate them into what a commander, analyst, policymaker, or defender should notice, care about, and do differently.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and

	cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 2. Lesson Outline

On-slide cues: Titan Rain - Strategic Espionage at Scale Estonia 2007 - Hostile Cyber Against a Nation Georgia 2008 - Cyber Meets Kinetic Stuxnet - The First Cyber Weapon The Escalation Pattern

Instructor talk track	Walk the class through the lesson arc from start to finish so they understand the narrative structure. Emphasize that each section builds toward an analytical framework rather than a list of isolated facts. Use the outline to show where you will define concepts, examine cases, and then apply them.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Use the outline to set pace expectations and tell students where you want questions held for discussion versus asked immediately.

Slide 3. Learning Objectives

On-slide cues: Explain Titan Rain as sustained strategic espionage Analyze Estonia 2007 as first attack on a nation-state Describe Georgia 2008 cyber-kinetic integration Explain why Stuxnet is the first cyber weapon Trace the escalation pattern: espionage -> destruction

Instructor talk track	Read the objectives as performance outcomes, not administrative text. Tell delegates that by the end of the session they must be able to explain, compare, and apply these concepts under discussion pressure. This helps convert passive listening into active analysis from the start.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Tell delegates exactly how they will prove mastery by the end of the session: define, compare, analyse, and apply.

Slide 4. Cyber History Timeline

On-slide cues: 1986-1998: Espionage emerges (Day 1 content) 2003-07: Titan Rain - Strategic espionage 2007: Estonia - National disruption 2008: Georgia - Hybrid warfare 2010: Stuxnet - Physical destruction

Instructor talk track	Use the timeline to show progression rather than isolated events. The slide works best when you narrate the shift from experimentation to espionage, then disruption, hybrid integration, and finally physical destruction.
Live internet demo / browse path	Use https://web.archive.org/ to show delegates how older web content can still be surfaced for historical analysis. This helps students see that cyber history is recoverable evidence, not just storytelling.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 5. Titan Rain

On-slide cues: 2003-2007: Strategic Espionage at Scale

Instructor talk track	Frame Titan Rain as strategic espionage at scale. The lesson is that cyber theft can erode a state's technological advantage over time without dramatic battlefield events, which makes strategic damage harder to see but still profound.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 6. Titan Rain - Overview

On-slide cues: Timeframe: 2003-2007 (possibly longer) Attribution: China (PLA Unit 61398 suspected) Targets: DoD, State Dept, DHS, contractors Method: Spearphishing, exploitation Data stolen: 10-20 terabytes (Library of Congress = 15TB)

Instructor talk	Frame Titan Rain as strategic espionage at scale. The lesson is that cyber theft

track	can erode a state's technological advantage over time without dramatic battlefield events, which makes strategic damage harder to see but still profound.
Live internet demo / browse path	Live browse a reputable search result for a public explainer on GPS jamming or radar spoofing, then contrast electromagnetic attack with pure network intrusion. Keep the demo conceptual and avoid any step-by-step tradecraft.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 7. Strategic Implications

On-slide cues: Cyber espionage can transfer tech advantage Years of R&D accessible to adversary Chinese military designs resemble US aircraft Cost: Billions in R&D stolen for minimal cost Prompted US Cyber Command discussions

Instructor talk track	Use this slide to show the maturation from isolated compromise to sustained state-backed espionage. The key teaching point is continuity: many modern APT patterns were visible early, and policy responses repeatedly lagged behind operational innovation.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 8. Estonia 2007

On-slide cues: Hostile Cyber Activity Against a Nation

Instructor talk track	Teach Estonia as a national-level disruption case where the technical attacks were less important than the political and cognitive effects. Stress that the operation exposed how a highly digitized society can experience coercion even without physical destruction or formally declared war.
Live internet demo / browse path	Open https://radar.cloudflare.com/outage-center or https://radar.cloudflare.com/ and demonstrate how modern outage or traffic-

	anomaly dashboards visualize disruption. Explain that Estonia predated these mature dashboards, but the logic of service disruption is the same.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 9. The Bronze Soldier Crisis

On-slide cues: Bronze Soldier: Soviet WWII memorial in Tallinn Estonian govt planned to relocate it Russian-speaking minority protested Moscow condemned the decision Riots erupted April 26-27, 2007

Instructor talk track	Teach Estonia as a national-level disruption case where the technical attacks were less important than the political and cognitive effects. Stress that the operation exposed how a highly digitized society can experience coercion even without physical destruction or formally declared war.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 10. The Cyber Campaign

On-slide cues: Phase 1 (Apr 27-29): Govt website defacement Phase 2 (Apr 30-May 9): Banks, media, telecom DDoS Phase 3 (May 9 - Victory Day): Peak attacks Targets: Parliament, ministries, banks, news Duration: Nearly 3 weeks of sustained attacks

Instructor talk track	Teach Estonia as a national-level disruption case where the technical attacks were less important than the political and cognitive effects. Stress that the operation exposed how a highly digitized society can experience coercion even without physical destruction or formally declared war.
Live internet demo / browse path	Open https://radar.cloudflare.com/outage-center or https://radar.cloudflare.com/ and demonstrate how modern outage or traffic-anomaly dashboards visualize disruption. Explain that Estonia predated these mature dashboards, but the logic of service disruption is the same.
Delegate action /	Pause and have delegates identify the key operational consequence of this

discussion move	slide before you move on.

Slide 11. Why Estonia Mattered

On-slide cues: First hostile cyber activity against nation-state First to trigger NATO Article 5 discussions First national-level cyber crisis Led to NATO CCDCOE in Tallinn Prompted Tallinn Manual development

Instructor talk track	Teach Estonia as a national-level disruption case where the technical attacks were less important than the political and cognitive effects. Stress that the operation exposed how a highly digitized society can experience coercion even without physical destruction or formally declared war.
Live internet demo / browse path	Open https://radar.cloudflare.com/outage-center or https://radar.cloudflare.com/ and demonstrate how modern outage or traffic-anomaly dashboards visualize disruption. Explain that Estonia predated these mature dashboards, but the logic of service disruption is the same.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 12. Georgia 2008

On-slide cues: Cyber Meets Kinetic Warfare

Instructor talk track	Use Georgia to illustrate cyber-kinetic integration. The critical insight is that cyber was not a side event; it supported the broader campaign by degrading communication, shaping information flow, and multiplying pressure during conventional operations.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 13. The Five-Day War

On-slide cues: Aug 7: Georgian forces enter South Ossetia Aug 8: Russian forces invade; cyber intensifies Aug 9-10: Russian advance + peak cyber attacks Aug 12: Ceasefire announced Cyber attacks continued days after ceasefire

Instructor talk track	Use Georgia to illustrate cyber-kinetic integration. The critical insight is that cyber was not a side event; it supported the broader campaign by degrading communication, shaping information flow, and multiplying pressure during conventional operations.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 14. What Made Georgia Different

On-slide cues: Estonia: Political dispute, no kinetic conflict Georgia: Active conventional warfare Estonia: Weeks of attacks Georgia: Days, synchronized with military ops Georgia: Cyber as military force multiplier

Instructor talk track	Use Georgia to illustrate cyber-kinetic integration. The critical insight is that cyber was not a side event; it supported the broader campaign by degrading communication, shaping information flow, and multiplying pressure during conventional operations.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 15. Lessons for Military Doctrine

On-slide cues: Cyber as force multiplier for conventional ops Information dominance is a military objective Timing and synchronization critical Deniability persists ('patriotic hackers') Hybrid warfare concept emerges

Instructor talk track	Use Georgia to illustrate cyber-kinetic integration. The critical insight is that cyber was not a side event; it supported the broader campaign by degrading communication, shaping information flow, and multiplying pressure during conventional operations.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 16. Stuxnet

On-slide cues: 2010: The First Cyber Weapon

Instructor talk track	Teach Stuxnet as the moment cyber crossed decisively into controlled physical sabotage. Emphasize the attack chain, the precision of target selection, the concealment from operators, and the strategic consequence that every state now has to take cyber-physical risk seriously.
Live internet demo / browse path	Browse https://www.cisa.gov/topics/industrial-control-systems and show the delegates how modern ICS and OT security is treated as national critical infrastructure security, not just traditional IT.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 17. Stuxnet - Discovery

On-slide cues: June 2010: Discovered by VirusBlokAda (Belarus) July 2010: Microsoft confirms 4 zero-day exploits Sept 2010: Symantec analysis reveals target Target: Iranian nuclear centrifuges Attribution: US/Israel joint op (never confirmed)

Instructor talk track	Teach Stuxnet as the moment cyber crossed decisively into controlled physical sabotage. Emphasize the attack chain, the precision of target selection, the concealment from operators, and the strategic consequence that every state now has to take cyber-physical risk seriously.
Live internet	Browse https://www.cisa.gov/topics/industrial-control-systems and show the

demo / browse path	delegates how modern ICS and OT security is treated as national critical infrastructure security, not just traditional IT.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 18. How Stuxnet Worked

On-slide cues: 1. USB infection into air-gapped network 2. Spread across Windows using zero-days 3. Searched for Siemens Step 7 software 4. Verified specific PLC configuration 5. Altered centrifuge speeds (sabotage) 6. Replayed normal readings (concealment)

Instructor talk track	Teach Stuxnet as the moment cyber crossed decisively into controlled physical sabotage. Emphasize the attack chain, the precision of target selection, the concealment from operators, and the strategic consequence that every state now has to take cyber-physical risk seriously.
Live internet demo / browse path	Browse https://www.cisa.gov/topics/industrial-control-systems and show the delegates how modern ICS and OT security is treated as national critical infrastructure security, not just traditional IT.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 19. Physical Destruction via Cyber

On-slide cues: ~1,000 Iranian centrifuges destroyed Nuclear program delayed 1-2 years No missiles, no troops, no casualties First cyber attack causing physical damage Proved cyber-physical effects are real

Instructor talk track	Teach Stuxnet as the moment cyber crossed decisively into controlled physical sabotage. Emphasize the attack chain, the precision of target selection, the concealment from operators, and the strategic consequence that every state now has to take cyber-physical risk seriously.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action /	Pause and have delegates identify the key operational consequence of this

discussion move	slide before you move on.

Slide 20. Implications and Legacy

On-slide cues: Cyber weapons are real - no longer theoretical Critical infrastructure is vulnerable Code leaked - proliferation concerns Legal/ethical questions raised Deterrence complicated (attribution problem)

Instructor talk track	Teach Stuxnet as the moment cyber crossed decisively into controlled physical sabotage. Emphasize the attack chain, the precision of target selection, the concealment from operators, and the strategic consequence that every state now has to take cyber-physical risk seriously.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 21. The Escalation Pattern

On-slide cues: Titan Rain (2003-07) -> ESPIONAGE at scale Estonia (2007) -> DISRUPTION of a nation Georgia (2008) -> INTEGRATION with warfare Stuxnet (2010) -> DESTRUCTION via cyber Each crossed a previously theoretical threshold

Instructor talk track	Use the timeline to show progression rather than isolated events. The slide works best when you narrate the shift from experimentation to espionage, then disruption, hybrid integration, and finally physical destruction.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 22. Key Takeaways

On-slide cues: Titan Rain: State espionage transfers tech advantage Estonia: Entire nations can be disrupted Georgia: Cyber integrates with conventional war Stuxnet: Cyber can cause physical destruction Escalation: Espionage -> Disruption -> Destruction

Instructor talk track	Use this slide to compress the lesson into five memory anchors. Re-state each point in plain language, then translate it into why it matters for commanders, analysts, or defenders. This is the moment to check whether students can move from vocabulary to judgement.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Ask delegates to summarize the slide using the formula: concept, risk, and military implication.

Slide 23. Preview: Day 3 Lab

On-slide cues: Apply MITRE ATT&CK to historical incidents Map techniques: Estonia, Georgia, or Stuxnet Compare techniques across incidents Identify defensive lessons Bring: ATT&CK Navigator access

Instructor talk track	Position the next class as a continuation of today's logic. Show delegates exactly how the next lesson will deepen or operationalize what they just learned so the course feels cumulative and coherent.
Live internet demo / browse path	Browse https://attack.mitre.org/matrices/enterprise/ and then launch the Navigator from https://attack.mitre.org/resources/attack-data-and-tools/. Show students how tactics, techniques, and heat maps will support the lab.
Delegate action / discussion move	Show them exactly what good output looks like so the next task feels executable, not vague.

Slide 24. Questions?

Instructor talk track	Use this slide as a deliberate debrief rather than a passive ending. Summarize the one idea students must carry forward from the session, restate the operational significance, and then invite analytical questions, not just clarification questions. Close by linking the current session to the next day so

	the storyline feels continuous.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Invite one operational question, one ethical question, and one technical question before closing.

Week 3 Day 1

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Use this section as your live instructor script. Each slide entry gives you a concise speaking flow, one internet-based illustration or demo path, and one delegate interaction or action to keep the room engaged.

#	Slide title	Primary facilitation angle
1	The Information Environment	Define the Information Environment broadly: people, organizations, and systems that create, move, and act on information.
2	Lesson Outline	Walk the class through the lesson arc from start to finish so they understand the narrative structure.
3	Learning Objectives	Read the objectives as performance outcomes, not administrative text.
4	The Information Environment	Define the Information Environment broadly: people, organizations, and systems that create, move, and act on information.
5	Defining the Information Environment	Define the Information Environment broadly: people, organizations, and systems that create, move, and act on information.
6	The Three Dimensions	Explain the three dimensions as nested layers through which influence and conflict operate.
7	Physical Dimension	Explain the three dimensions as nested layers through which influence and conflict operate.
8	https://www.submarinecablemap.com/	Use this slide to explain the operational meaning of the listed points: .
9	Informational Dimension	Explain the three dimensions as nested layers through which influence and conflict operate.
10	Cognitive Dimension	Explain the three dimensions as nested layers through which influence and conflict operate.
11	The Cognitive Dimension	Explain the three dimensions as nested layers through which influence and conflict operate.
12	Why Cognition Matters Most	Drive home that will, legitimacy, trust, and

		perception often decide conflicts before annihilation does.
13	Historical Example: Cold War	Drive home that will, legitimacy, trust, and perception often decide conflicts before annihilation does.
14	Information as a Weapon	Drive home that will, legitimacy, trust, and perception often decide conflicts before annihilation does.
15	Connecting Cyber to Information	Teach this slide as an analytical bridge.
16	How Cyber Affects the IE	Teach this slide as an analytical bridge.
17	Estonia 2007 Through IE Lens	Teach Estonia as a national-level disruption case where the technical attacks were less important than the political and cognitive effects.
18	Hack-and-Leak Operations	Teach this slide as an analytical bridge.
19	The Amplification Problem	Teach this slide as an analytical bridge.
20	Implications and Framework	Use this slide to explain the operational meaning of the listed points: Applying What We've Learned.
21	Why This Framework Matters	Teach this slide as an analytical bridge.
22	Analysis Framework	Teach this slide as an analytical bridge.
23	Key Takeaways	Use this slide to compress the lesson into five memory anchors.
24	Preview: Day 2	Position the next class as a continuation of today's logic.
25	Questions?	Use this slide as a deliberate debrief rather than a passive ending.

Slide 1. The Information Environment

On-slide cues: History and Trends of Warfare in the Fifth Domain

Instructor talk track	Define the Information Environment broadly: people, organizations, and systems that create, move, and act on information. Make the point that cyber is only one part of this bigger environment and that modern conflict is increasingly about shaping what actors perceive and decide.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and

	cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 2. Lesson Outline

On-slide cues: What is the Information Environment? The Three Dimensions of the IE The Cognitive Dimension — Ultimate Target Connecting Cyber to Information Implications and Framework

Instructor talk track	Walk the class through the lesson arc from start to finish so they understand the narrative structure. Emphasize that each section builds toward an analytical framework rather than a list of isolated facts. Use the outline to show where you will define concepts, examine cases, and then apply them.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Use the outline to set pace expectations and tell students where you want questions held for discussion versus asked immediately.

Slide 3. Learning Objectives

On-slide cues: Define the Information Environment and its significance Describe the three dimensions: physical, informational, cognitive Explain how the cognitive dimension is the ultimate target Analyze relationship between cyber and the IE Identify historical examples of information as a weapon

Instructor talk track	Read the objectives as performance outcomes, not administrative text. Tell delegates that by the end of the session they must be able to explain, compare, and apply these concepts under discussion pressure. This helps convert passive listening into active analysis from the start.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.

Delegate action / discussion move	Tell delegates exactly how they will prove mastery by the end of the session: define, compare, analyse, and apply.

Slide 4. The Information Environment

On-slide cues: Where Modern Conflicts Are Fought

Instructor talk track	Define the Information Environment broadly: people, organizations, and systems that create, move, and act on information. Make the point that cyber is only one part of this bigger environment and that modern conflict is increasingly about shaping what actors perceive and decide.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 5. Defining the Information Environment

On-slide cues: JP 3-13: “Aggregate of individuals, organizations, and systems that collect, process, disseminate, or act on information” Individuals — Decision-makers, populations, analysts Organizations — Governments, media, corporations Systems — Networks, databases, platforms, broadcast

Instructor talk track	Define the Information Environment broadly: people, organizations, and systems that create, move, and act on information. Make the point that cyber is only one part of this bigger environment and that modern conflict is increasingly about shaping what actors perceive and decide.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 6. The Three Dimensions

On-slide cues: PHYSICAL Hardware, infrastructure, tangible systems | INFORMATIONAL Data, content, messages, protocols | COGNITIVE Human minds, perceptions, decisions | The Cognitive Dimension is the Ultimate Target | Nested structure: Each layer depends on the others. Effects cascade across all three dimensions.

Instructor talk track	Explain the three dimensions as nested layers through which influence and conflict operate. Students should leave this slide understanding that physical infrastructure and information content matter, but the decisive end state usually appears in the cognitive dimension where people judge, choose, and act.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 7. Physical Dimension

On-slide cues: Data centers, undersea cables, cell towers Servers, computers, radios, satellites Broadcast studios, printing facilities Geographic locations of infrastructure Key insight: Can be physically destroyed or degraded

Instructor talk track	Explain the three dimensions as nested layers through which influence and conflict operate. Students should leave this slide understanding that physical infrastructure and information content matter, but the decisive end state usually appears in the cognitive dimension where people judge, choose, and act.
Live internet demo / browse path	Browse https://www.submarinecablemap.com/ and zoom into Africa or Europe. Show that cyberspace still depends on physical chokepoints, landing stations, and owned infrastructure.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 8. https://www.submarinecablemap.com/

Instructor talk track	Use this slide to explain the operational meaning of the listed points: . Do not just read the bullets. Translate them into what a commander, analyst, policymaker, or defender should notice, care about, and do differently.
Live internet demo / browse path	Browse https://www.submarinecablemap.com/ and zoom into Africa or Europe. Show that cyberspace still depends on physical chokepoints, landing stations, and owned infrastructure.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 9. Informational Dimension

On-slide cues: Data and content (databases, news, posts) Protocols and formats (how info is structured) Collection and storage mechanisms Where information flows between systems Key insight: Can be manipulated, altered, deleted, leaked

Instructor talk track	Explain the three dimensions as nested layers through which influence and conflict operate. Students should leave this slide understanding that physical infrastructure and information content matter, but the decisive end state usually appears in the cognitive dimension where people judge, choose, and act.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 10. Cognitive Dimension

On-slide cues: Human perception and interpretation Decision-making processes Beliefs, values, and worldviews Emotions: fear, anger, trust, doubt ▶️ The ULTIMATE TARGET of information warfare

Instructor talk track	Explain the three dimensions as nested layers through which influence and conflict operate. Students should leave this slide understanding that physical infrastructure and information content matter, but the decisive end state usually appears in the cognitive dimension where people judge, choose, and

	act.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 11. The Cognitive Dimension

On-slide cues: Where Wars Are Won and Lost

Instructor talk track	Explain the three dimensions as nested layers through which influence and conflict operate. Students should leave this slide understanding that physical infrastructure and information content matter, but the decisive end state usually appears in the cognitive dimension where people judge, choose, and act.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 12. Why Cognition Matters Most

On-slide cues: Wars are won through will and decision-making Physical destruction is means, not end Adversary decides when they've lost Public opinion constrains/enables action Leaders make decisions based on perceptions

Instructor talk track	Drive home that will, legitimacy, trust, and perception often decide conflicts before annihilation does. Use the Cold War example to show that cognitive victory can collapse an adversary's model of reality without requiring battlefield conquest.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and

	cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 13. Historical Example: Cold War

On-slide cues: Fundamentally an information war Both sides had nuclear arsenals (MAD) Conflict decided in cognitive dimension US: Freedom, prosperity, democracy USSR: Equality, anti-imperialism Soviet system lost internal legitimacy

Instructor talk track	Drive home that will, legitimacy, trust, and perception often decide conflicts before annihilation does. Use the Cold War example to show that cognitive victory can collapse an adversary's model of reality without requiring battlefield conquest.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 14. Information as a Weapon

On-slide cues: Actions to achieve information superiority Affect adversary information and decision-making While defending your own The Cognitive Kill Chain: Find -> Fix -> Track -> Target -> Engage -> Assess

Instructor talk track	Drive home that will, legitimacy, trust, and perception often decide conflicts before annihilation does. Use the Cold War example to show that cognitive victory can collapse an adversary's model of reality without requiring battlefield conquest.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.

Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 15. Connecting Cyber to Information

On-slide cues: How Cyber Operations Affect the IE

Instructor talk track	Teach this slide as an analytical bridge. Cyber can act through, against, and within the information environment, but the meaningful question is always what perception or decision the technical action was meant to shape.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 16. How Cyber Affects the IE

On-slide cues: THROUGH the IE — Use cyber to deliver information Example: Hack-and-leak operations AGAINST the IE — Disrupt information systems Example: DDoS on news websites WITHIN the IE — Operate in informational dimension Example: Social media manipulation

Instructor talk track	Teach this slide as an analytical bridge. Cyber can act through, against, and within the information environment, but the meaningful question is always what perception or decision the technical action was meant to shape.
Live internet demo / browse path	Open https://radar.cloudflare.com/outage-center or https://radar.cloudflare.com/ and demonstrate how modern outage or traffic-anomaly dashboards visualize disruption. Explain that Estonia predated these mature dashboards, but the logic of service disruption is the same.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 17. Estonia 2007 Through IE Lens

On-slide cues: PHYSICAL: Servers overwhelmed, infrastructure stressed INFORMATIONAL: Govt websites down, news disrupted COGNITIVE: Fear, uncertainty, doubt in government Technical attacks were unsophisticated Cognitive effect was the real objective

Instructor talk track	Teach Estonia as a national-level disruption case where the technical attacks were less important than the political and cognitive effects. Stress that the operation exposed how a highly digitized society can experience coercion even without physical destruction or formally declared war.
Live internet demo / browse path	Open https://radar.cloudflare.com/outage-center or https://radar.cloudflare.com/ and demonstrate how modern outage or traffic-anomaly dashboards visualize disruption. Explain that Estonia predated these mature dashboards, but the logic of service disruption is the same.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 18. Hack-and-Leak Operations

On-slide cues: Gain access to target systems (cyber) Exfiltrate sensitive information Time release for maximum cognitive impact Use cutouts/proxies for disclosure Target's own information destroys credibility

Instructor talk track	Teach this slide as an analytical bridge. Cyber can act through, against, and within the information environment, but the meaningful question is always what perception or decision the technical action was meant to shape.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 19. The Amplification Problem

On-slide cues: Reach Limited channels -> Anyone can broadcast globally Speed Hours to days -> Seconds Gatekeeping Editors, censors -> Algorithms Persistence Ephemeral -> Permanent archive

Instructor talk track	Teach this slide as an analytical bridge. Cyber can act through, against, and within the information environment, but the meaningful question is always what perception or decision the technical action was meant to shape.
Live internet demo / browse path	Browse https://euvsdisinfo.eu/ and show a current example of coordinated narrative framing. Ask delegates to distinguish between narrative, channel, amplifier, and intended cognitive effect.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 20. Implications and Framework

On-slide cues: Applying What We've Learned

Instructor talk track	Use this slide to explain the operational meaning of the listed points: Applying What We've Learned. Do not just read the bullets. Translate them into what a commander, analyst, policymaker, or defender should notice, care about, and do differently.
Live internet demo / browse path	Live browse a reputable search result for a public explainer on GPS jamming or radar spoofing, then contrast electromagnetic attack with pure network intrusion. Keep the demo conceptual and avoid any step-by-step tradecraft.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 21. Why This Framework Matters

On-slide cues: FOR MILITARY: Every operation has information effects FOR DEFENSE: Must protect all three dimensions FOR ANALYSIS: Ask “what was the cognitive objective?” FOR STRATEGY: Technical effects are often means, not ends FOR POLICY: Cannot separate cyber from information

Instructor talk track	Teach this slide as an analytical bridge. Cyber can act through, against, and within the information environment, but the meaningful question is always what perception or decision the technical action was meant to shape.
Live internet demo / browse path	Live browse a reputable search result for a public explainer on GPS jamming or radar spoofing, then contrast electromagnetic attack with pure network

	intrusion. Keep the demo conceptual and avoid any step-by-step tradecraft.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 22. Analysis Framework

On-slide cues: What physical systems were affected? What information was accessed/modified/disclosed? What cognitive effect was intended/achieved? How do the three dimensions interact? What was the strategic objective?

Instructor talk track	Teach this slide as an analytical bridge. Cyber can act through, against, and within the information environment, but the meaningful question is always what perception or decision the technical action was meant to shape.
Live internet demo / browse path	Live browse a reputable search result for a public explainer on GPS jamming or radar spoofing, then contrast electromagnetic attack with pure network intrusion. Keep the demo conceptual and avoid any step-by-step tradecraft.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 23. Key Takeaways

On-slide cues: IE = individuals, organizations, systems + information Three dimensions: Physical, Informational, Cognitive Cognitive dimension is the ultimate target Cyber operations affect all three dimensions Modern amplification makes cognitive warfare potent

Instructor talk track	Use this slide to compress the lesson into five memory anchors. Re-state each point in plain language, then translate it into why it matters for commanders, analysts, or defenders. This is the moment to check whether students can move from vocabulary to judgement.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Ask delegates to summarize the slide using the formula: concept, risk, and military implication.

Slide 24. Preview: Day 2

On-slide cues: Information Operations — deliberate actions in the IE Electronic Warfare (EW) — controlling EM spectrum Psychological Operations (MISO) — influencing audiences Cyber Operations — integration with EW and MISO Real-world campaigns — Russian IO, extremist propaganda

Instructor talk track	Position the next class as a continuation of today's logic. Show delegates exactly how the next lesson will deepen or operationalize what they just learned so the course feels cumulative and coherent.
Live internet demo / browse path	Live browse a reputable search result for a public explainer on GPS jamming or radar spoofing, then contrast electromagnetic attack with pure network intrusion. Keep the demo conceptual and avoid any step-by-step tradecraft.
Delegate action / discussion move	Show them exactly what good output looks like so the next task feels executable, not vague.

Slide 25. Questions?

Instructor talk track	Use this slide as a deliberate debrief rather than a passive ending. Summarize the one idea students must carry forward from the session, restate the operational significance, and then invite analytical questions, not just clarification questions. Close by linking the current session to the next day so the storyline feels continuous.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Invite one operational question, one ethical question, and one technical question before closing.

Week 3 Day 2

Deck file: HW401_Week3_Day2_Presentation.pptx | 25 slides

Use this section as your live instructor script. Each slide entry gives you a concise speaking flow, one internet-based illustration or demo path, and one delegate interaction or action to keep the room engaged.

#	Slide title	Primary facilitation angle
1	Information Operations	Explain Information Operations as an integrated toolkit for shaping adversary decision-making while protecting our own.
2	Lesson Outline	Walk the class through the lesson arc from start to finish so they understand the narrative structure.
3	Learning Objectives	Read the objectives as performance outcomes, not administrative text.
4	Defining Information Operations	Explain Information Operations as an integrated toolkit for shaping adversary decision-making while protecting our own.
5	What are Information Operations?	Explain Information Operations as an integrated toolkit for shaping adversary decision-making while protecting our own.
6	The IO Toolkit	Explain Information Operations as an integrated toolkit for shaping adversary decision-making while protecting our own.
7	IO vs. Information Warfare	Explain Information Operations as an integrated toolkit for shaping adversary decision-making while protecting our own.
8	Electronic Warfare	Teach EW as control of the electromagnetic spectrum and then show why that matters to cyber-enabled operations.
9	Electronic Warfare Fundamentals	Teach EW as control of the electromagnetic spectrum and then show why that matters to cyber-enabled operations.
10	EW and Cyber Convergence	Teach EW as control of the electromagnetic spectrum and then show why that matters to cyber-enabled operations.
11	EW in Practice: Operation Orchard	Teach EW as control of the electromagnetic spectrum and then show why that matters to cyber-enabled operations.

12	Psychological Operations	Explain MISO as targeted influence aimed at behavior change, not vague persuasion.
13	MISO Fundamentals	Explain MISO as targeted influence aimed at behavior change, not vague persuasion.
14	Historical MISO Examples	Explain MISO as targeted influence aimed at behavior change, not vague persuasion.
15	Modern MISO Challenges	Explain MISO as targeted influence aimed at behavior change, not vague persuasion.
16	Integration and Modern Campaigns	Use this slide to show the multiplication effect of combined capabilities.
17	The Integration Model	Use this slide to show the multiplication effect of combined capabilities.
18	Integration Example Scenario	Use this slide to show the multiplication effect of combined capabilities.
19	Russian Information Warfare	Teach this content as a live problem set, not distant theory.
20	Internet Research Agency (IRA)	Teach this content as a live problem set, not distant theory.
21	Disinformation vs. Misinformation	Teach this content as a live problem set, not distant theory.
22	Emerging Threat: Deepfakes	Teach this content as a live problem set, not distant theory.
23	Key Takeaways	Use this slide to compress the lesson into five memory anchors.
24	Preview: Day 3 Lab	Position the next class as a continuation of today's logic.
25	Questions?	Use this slide as a deliberate debrief rather than a passive ending.

Slide 1. Information Operations

On-slide cues: History and Trends of Warfare in the Fifth Domain

Instructor talk track	Explain Information Operations as an integrated toolkit for shaping adversary decision-making while protecting our own. Use the slide to contrast doctrinal language, especially the difference between a Western war-time framing and the Russian idea of continuous information confrontation.
Live internet	Use a quick live browse to anchor one current, real-world example connected

demo / browse path	to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 2. Lesson Outline

On-slide cues: Defining Information Operations Electronic Warfare (EW) Fundamentals Psychological Operations / MISO Integration and Modern Campaigns Russian Information Warfare Case Study

Instructor talk track	Walk the class through the lesson arc from start to finish so they understand the narrative structure. Emphasize that each section builds toward an analytical framework rather than a list of isolated facts. Use the outline to show where you will define concepts, examine cases, and then apply them.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Use the outline to set pace expectations and tell students where you want questions held for discussion versus asked immediately.

Slide 3. Learning Objectives

On-slide cues: Define Information Operations and its components Explain Electronic Warfare concepts Describe Psychological Operations (MISO) Analyze how EW, PsyOps, and Cyber integrate Examine real-world IO campaigns

Instructor talk track	Read the objectives as performance outcomes, not administrative text. Tell delegates that by the end of the session they must be able to explain, compare, and apply these concepts under discussion pressure. This helps convert passive listening into active analysis from the start.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action /	Tell delegates exactly how they will prove mastery by the end of the session:

discussion move	define, compare, analyse, and apply.

Slide 4. Defining Information Operations

On-slide cues: The Toolkit for the Information Environment

Instructor talk track	Explain Information Operations as an integrated toolkit for shaping adversary decision-making while protecting our own. Use the slide to contrast doctrinal language, especially the difference between a Western war-time framing and the Russian idea of continuous information confrontation.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 5. What are Information Operations?

On-slide cues: JP 3-13: 'Integrated employment of information-related capabilities to influence, disrupt, corrupt, or usurp adversary decision-making' Target: Decision-making processes Protecting our own decision-making

Instructor talk track	Explain Information Operations as an integrated toolkit for shaping adversary decision-making while protecting our own. Use the slide to contrast doctrinal language, especially the difference between a Western war-time framing and the Russian idea of continuous information confrontation.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 6. The IO Toolkit

On-slide cues: CORE CAPABILITIES: Electronic Warfare (EW) Psychological Operations (MISO) Cyber Operations SUPPORTING CAPABILITIES: OPSEC, Military Deception, Public Affairs

Instructor talk track	Explain Information Operations as an integrated toolkit for shaping adversary decision-making while protecting our own. Use the slide to contrast doctrinal language, especially the difference between a Western war-time framing and the Russian idea of continuous information confrontation.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 7. IO vs. Information Warfare

On-slide cues: US Doctrine: IO during military operations Russian Doctrine: Information warfare is constant US: War/peace distinction Russia: 'Information confrontation' always ongoing This difference has strategic implications

Instructor talk track	Explain Information Operations as an integrated toolkit for shaping adversary decision-making while protecting our own. Use the slide to contrast doctrinal language, especially the difference between a Western war-time framing and the Russian idea of continuous information confrontation.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 8. Electronic Warfare

On-slide cues: Controlling the Electromagnetic Spectrum

Instructor talk track	Teach EW as control of the electromagnetic spectrum and then show why that matters to cyber-enabled operations. Wireless communications, sensors, navigation, and radar all live in this contested space, so modern EW and cyber planning increasingly overlap.
Live internet	Live browse a reputable search result for a public explainer on GPS jamming

demo / browse path	or radar spoofing, then contrast electromagnetic attack with pure network intrusion. Keep the demo conceptual and avoid any step-by-step tradecraft.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 9. Electronic Warfare Fundamentals

On-slide cues: Definition: Military action to control EM spectrum THREE PILLARS: Electronic Attack (EA) - Deny adversary use Electronic Protection (EP) - Ensure friendly use EW Support (ES) - Detect and locate emissions

Instructor talk track	Teach EW as control of the electromagnetic spectrum and then show why that matters to cyber-enabled operations. Wireless communications, sensors, navigation, and radar all live in this contested space, so modern EW and cyber planning increasingly overlap.
Live internet demo / browse path	Live browse a reputable search result for a public explainer on GPS jamming or radar spoofing, then contrast electromagnetic attack with pure network intrusion. Keep the demo conceptual and avoid any step-by-step tradecraft.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 10. EW and Cyber Convergence

On-slide cues: EM spectrum carries wireless cyber operations Wi-Fi, cellular, satellite = EW and cyber overlap Jamming GPS = cyber-physical effects Signal spoofing = deception through spectrum SIGINT collection = intercepting data

Instructor talk track	Teach EW as control of the electromagnetic spectrum and then show why that matters to cyber-enabled operations. Wireless communications, sensors, navigation, and radar all live in this contested space, so modern EW and cyber planning increasingly overlap.
Live internet demo / browse path	Live browse a reputable search result for a public explainer on GPS jamming or radar spoofing, then contrast electromagnetic attack with pure network intrusion. Keep the demo conceptual and avoid any step-by-step tradecraft.
Delegate action /	Pause and have delegates identify the key operational consequence of this

discussion move	slide before you move on.

Slide 11. EW in Practice: Operation Orchard

On-slide cues: 2007: Israel strikes Syrian reactor Syrian air defenses were sophisticated Israeli aircraft 'blinded' Syrian radar Operators saw nothing as aircraft entered EW enabled kinetic strike success

Instructor talk track	Teach EW as control of the electromagnetic spectrum and then show why that matters to cyber-enabled operations. Wireless communications, sensors, navigation, and radar all live in this contested space, so modern EW and cyber planning increasingly overlap.
Live internet demo / browse path	Live browse a reputable search result for a public explainer on GPS jamming or radar spoofing, then contrast electromagnetic attack with pure network intrusion. Keep the demo conceptual and avoid any step-by-step tradecraft.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 12. Psychological Operations

On-slide cues: Military Information Support Operations (MISO)

Instructor talk track	Explain MISO as targeted influence aimed at behavior change, not vague persuasion. Use the historical examples to show continuity of purpose and then contrast older one-way channels with today's algorithmic, interactive, and much harder-to-control information ecosystem.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 13. MISO Fundamentals

On-slide cues: Planned operations to convey information Target: Foreign audiences Goal: Influence emotions, motives, decisions Ultimately: Influence behavior US MISO must be based on truth

Instructor talk track	Explain MISO as targeted influence aimed at behavior change, not vague persuasion. Use the historical examples to show continuity of purpose and then contrast older one-way channels with today's algorithmic, interactive, and much harder-to-control information ecosystem.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 14. Historical MISO Examples

On-slide cues: WWII: Operation Cornflakes - fake mail to Germans Vietnam: Chieu Hoi - 200,000 defectors via leaflets Gulf War: 29 million leaflets, mass surrenders Methods evolved: leaflets -> radio -> social media Core principle unchanged: influence decisions

Instructor talk track	Explain MISO as targeted influence aimed at behavior change, not vague persuasion. Use the historical examples to show continuity of purpose and then contrast older one-way channels with today's algorithmic, interactive, and much harder-to-control information ecosystem.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 15. Modern MISO Challenges

On-slide cues: PAST: Limited channels (radio, leaflets) PRESENT: Infinite channels (social media) PAST: One-way broadcast PRESENT: Two-way engagement, counter-messaging PRESENT: Algorithms determine reach, not operators

Instructor talk track	Explain MISO as targeted influence aimed at behavior change, not vague persuasion. Use the historical examples to show continuity of purpose and then contrast older one-way channels with today's algorithmic, interactive,

	and much harder-to-control information ecosystem.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 16. Integration and Modern Campaigns

On-slide cues: How EW, MISO, and Cyber Work Together

Instructor talk track	Use this slide to show the multiplication effect of combined capabilities. Cyber can provide access, EW can deny verification, and MISO can shape the content; together they create a cognitive outcome far stronger than any one capability could create alone.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 17. The Integration Model

On-slide cues: All three capabilities target cognitive effects EW: Controls spectrum -> affects communications MISO: Shapes perceptions -> affects decisions Cyber: Enables both + direct effects Integration multiplies effectiveness

Instructor talk track	Use this slide to show the multiplication effect of combined capabilities. Cyber can provide access, EW can deny verification, and MISO can shape the content; together they create a cognitive outcome far stronger than any one capability could create alone.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.

Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 18. Integration Example Scenario

On-slide cues: CYBER: Compromise enemy C2 network MISO: Send false orders via compromised system EW: Jam radio frequencies RESULT: Enemy receives false orders, can't verify COGNITIVE EFFECT: Command paralysis

Instructor talk track	Use this slide to show the multiplication effect of combined capabilities. Cyber can provide access, EW can deny verification, and MISO can shape the content; together they create a cognitive outcome far stronger than any one capability could create alone.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 19. Russian Information Warfare

On-slide cues: Western approach: IO during conflict Russian approach: Continuous, including peacetime Western targets: Military decision-makers Russian targets: Entire populations Western: MISO must be truthful Russian: 'Truth is what serves the state'

Instructor talk track	Teach this content as a live problem set, not distant theory. Stress continuous competition, the use of fake personas and narrative amplification, the difference between disinformation and misinformation, and the way synthetic media now attacks trust itself.
Live internet demo / browse path	Browse https://euvsdisinfo.eu/ and show a current example of coordinated narrative framing. Ask delegates to distinguish between narrative, channel, amplifier, and intended cognitive effect.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 20. Internet Research Agency (IRA)

On-slide cues: Russian organization conducting influence ops Hundreds of employees, 24/7 operations Fake social media personas across platforms Divisive content on ALL sides of issues Goal: Amplify existing divisions, not create new

Instructor talk track	Teach this content as a live problem set, not distant theory. Stress continuous competition, the use of fake personas and narrative amplification, the difference between disinformation and misinformation, and the way synthetic media now attacks trust itself.
Live internet demo / browse path	Browse https://euvsdisinfo.eu/ and show a current example of coordinated narrative framing. Ask delegates to distinguish between narrative, channel, amplifier, and intended cognitive effect.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 21. Disinformation vs. Misinformation

On-slide cues: DISINFORMATION: Deliberately false, intent to deceive MISINFORMATION: False, but no deceptive intent MALINFORMATION: True, but used maliciously Disinfo becomes misinfo when shared by believers This is why attribution and education matter

Instructor talk track	Teach this content as a live problem set, not distant theory. Stress continuous competition, the use of fake personas and narrative amplification, the difference between disinformation and misinformation, and the way synthetic media now attacks trust itself.
Live internet demo / browse path	Browse https://euvsdisinfo.eu/ and show a current example of coordinated narrative framing. Ask delegates to distinguish between narrative, channel, amplifier, and intended cognitive effect.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 22. Emerging Threat: Deepfakes

On-slide cues: AI-generated synthetic media Audio and video can be fabricated Increasing realism, decreasing cost Challenge: 'Seeing is believing' no longer true Implications for evidence, trust, attribution

Instructor talk track	Teach this content as a live problem set, not distant theory. Stress continuous competition, the use of fake personas and narrative amplification, the difference between disinformation and misinformation, and the way synthetic media now attacks trust itself.
Live internet demo / browse path	Open https://www.cisa.gov/news-events/alerts/2023/09/12/nsa-fbi-and-cisa-release-cybersecurity-information-sheet-deepfake-threats and show that synthetic media risk is now addressed by security agencies as an operational threat to trust, identity, and crisis response.
Delegate action / discussion move	Pause and have delegates identify the key operational consequence of this slide before you move on.

Slide 23. Key Takeaways

On-slide cues: IO = integrated employment of EW, MISO, Cyber EW controls EM spectrum (attack, protect, support) MISO influences foreign audiences (truthfully in US) Integration multiplies effects toward cognitive goals Russian IO differs: continuous, targets populations

Instructor talk track	Use this slide to compress the lesson into five memory anchors. Re-state each point in plain language, then translate it into why it matters for commanders, analysts, or defenders. This is the moment to check whether students can move from vocabulary to judgement.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Ask delegates to summarize the slide using the formula: concept, risk, and military implication.

Slide 24. Preview: Day 3 Lab

On-slide cues: Analyze real IO campaigns using framework Identify indicators of coordinated behavior Map techniques to IE dimensions Conduct OSINT research on IO campaigns Present findings to class

Instructor talk track	Position the next class as a continuation of today's logic. Show delegates exactly how the next lesson will deepen or operationalize what they just

	learned so the course feels cumulative and coherent.
Live internet demo / browse path	Browse https://attack.mitre.org/matrices/enterprise/ and then launch the Navigator from https://attack.mitre.org/resources/attack-data-and-tools/. Show students how tactics, techniques, and heat maps will support the lab.
Delegate action / discussion move	Show them exactly what good output looks like so the next task feels executable, not vague.

Slide 25. Questions?

Instructor talk track	Use this slide as a deliberate debrief rather than a passive ending. Summarize the one idea students must carry forward from the session, restate the operational significance, and then invite analytical questions, not just clarification questions. Close by linking the current session to the next day so the storyline feels continuous.
Live internet demo / browse path	Use a quick live browse to anchor one current, real-world example connected to this slide. Ask delegates to identify the physical, informational, and cognitive elements they can observe in the source.
Delegate action / discussion move	Invite one operational question, one ethical question, and one technical question before closing.